

Computer Vision, Color

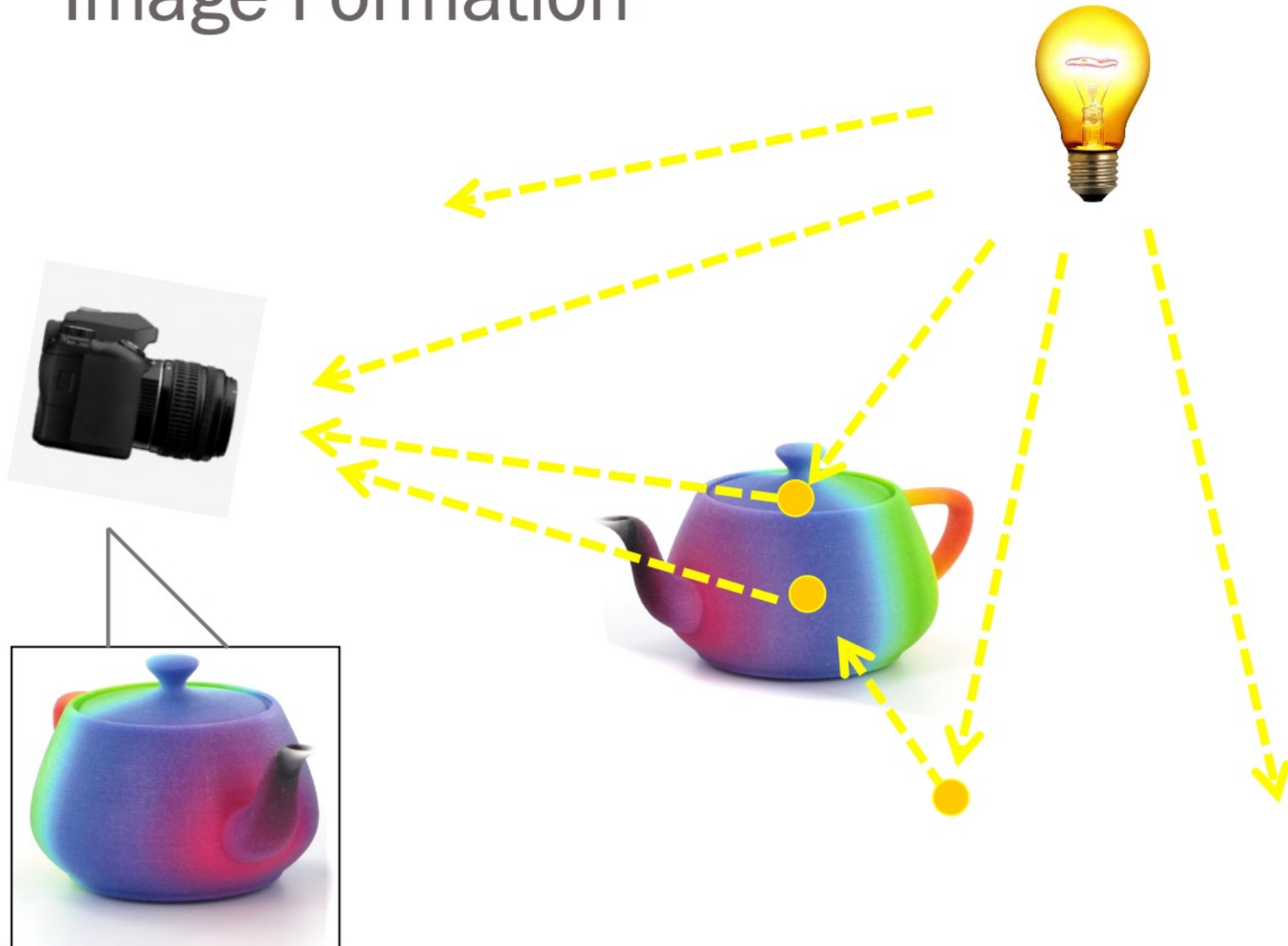
Darius Burschka

Color

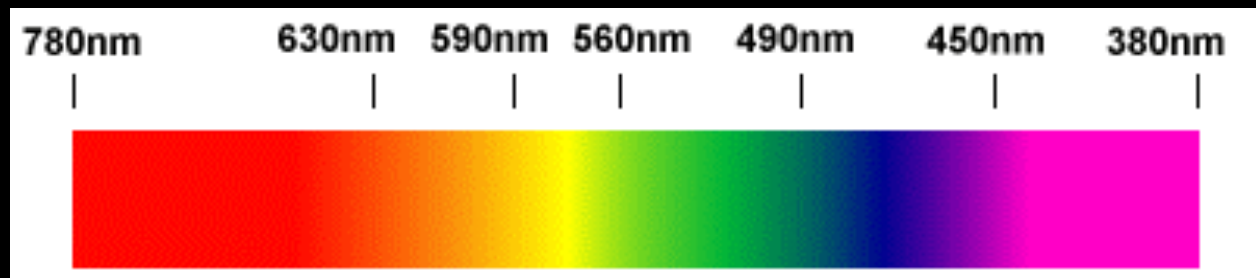
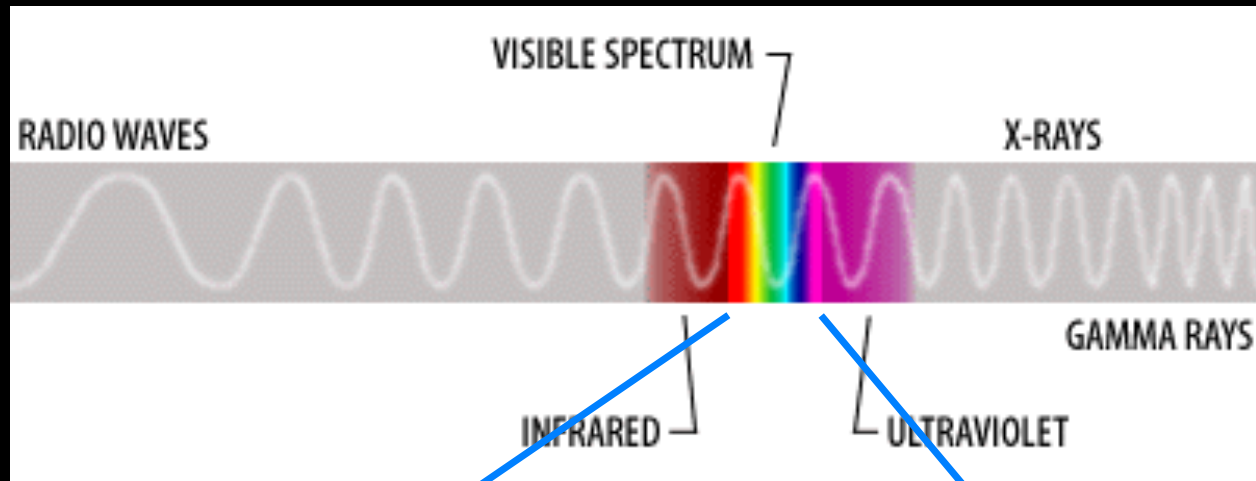
- Questions
 - What are the physics of color?
 - How is color sensed?
 - How is it represented?



Image Formation

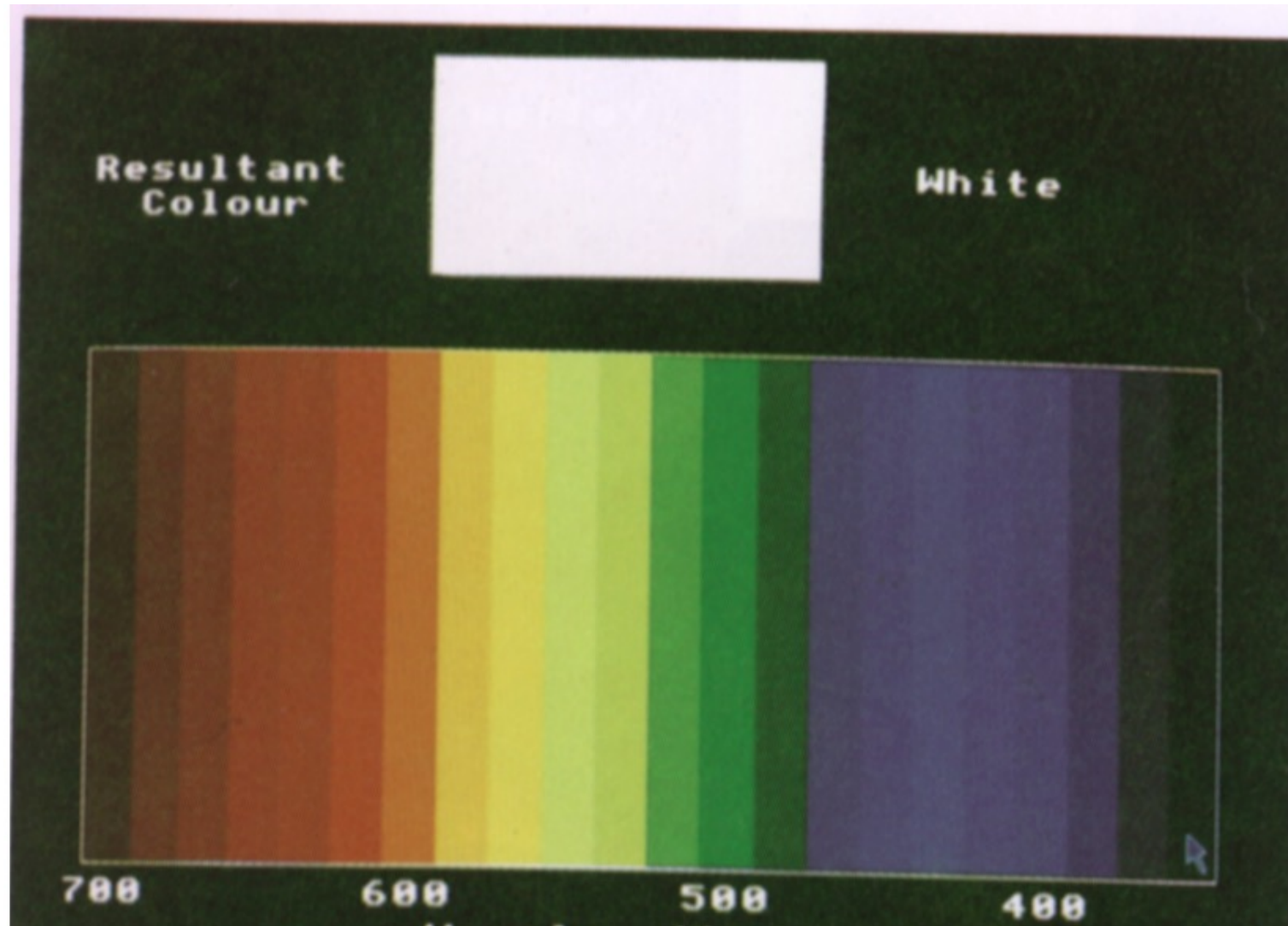


Electromagnetic Spectrum

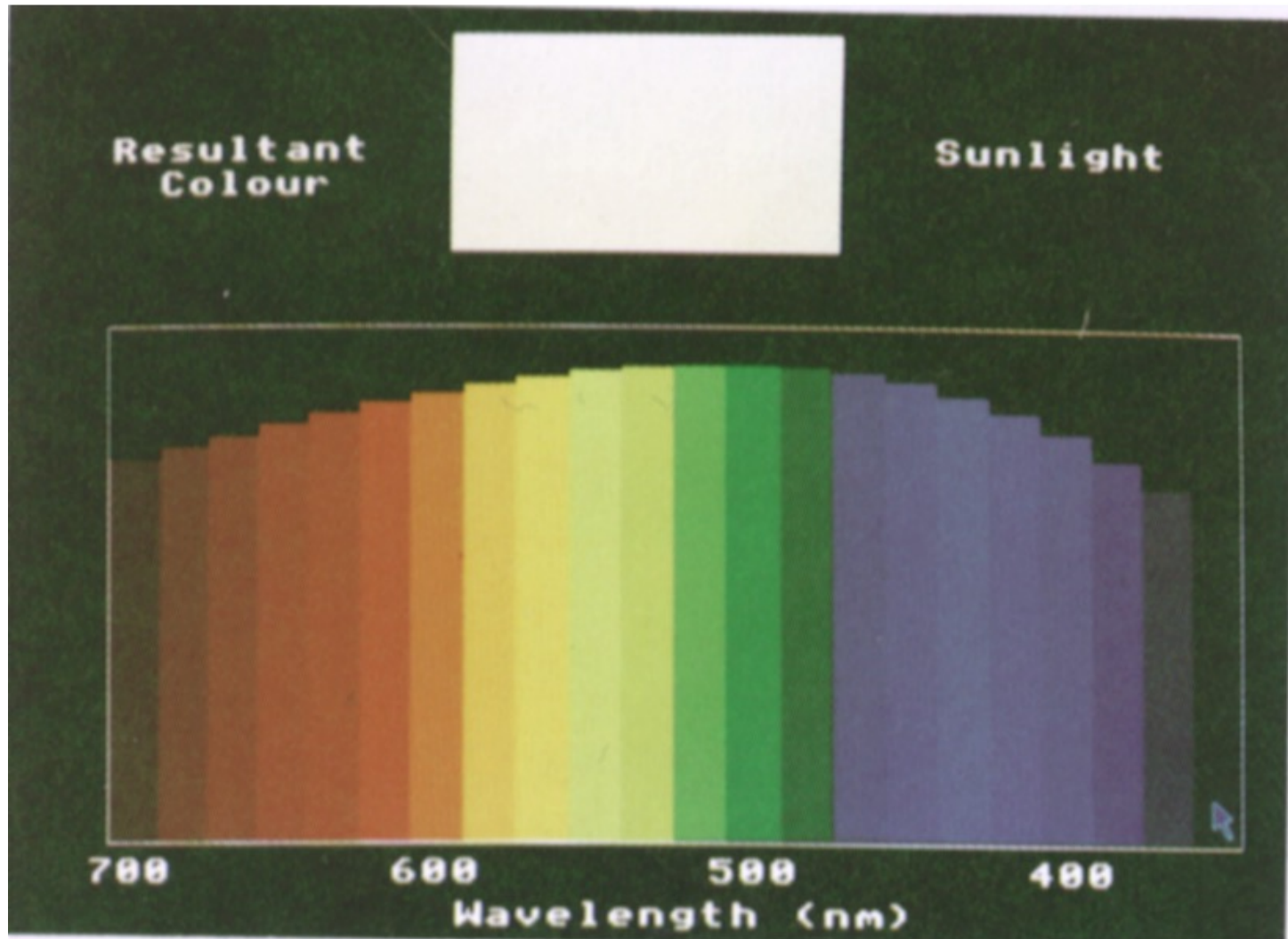


‘Visible’ Spectrum

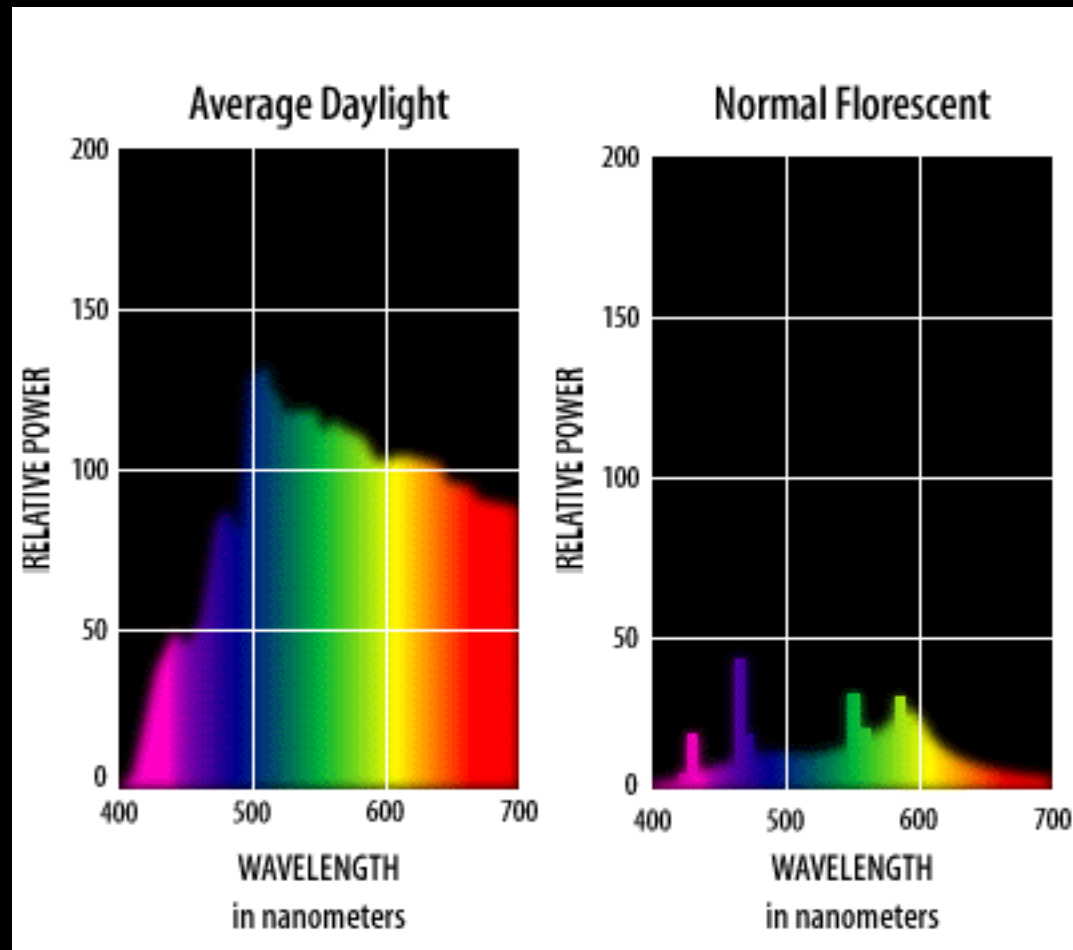
Light Spectra



Sunlight

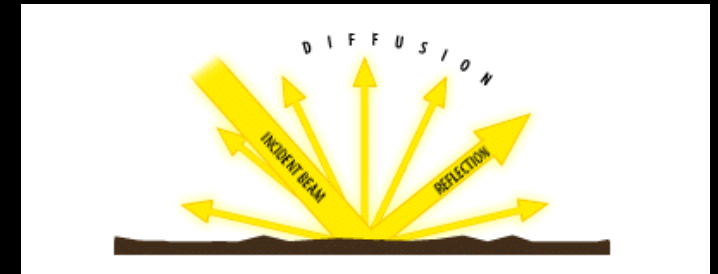
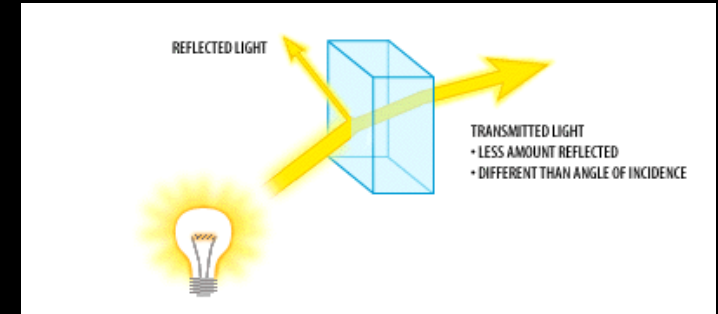


- Spectral distributions show the ‘amount’ of energy at each wavelength for a light source; e.g.



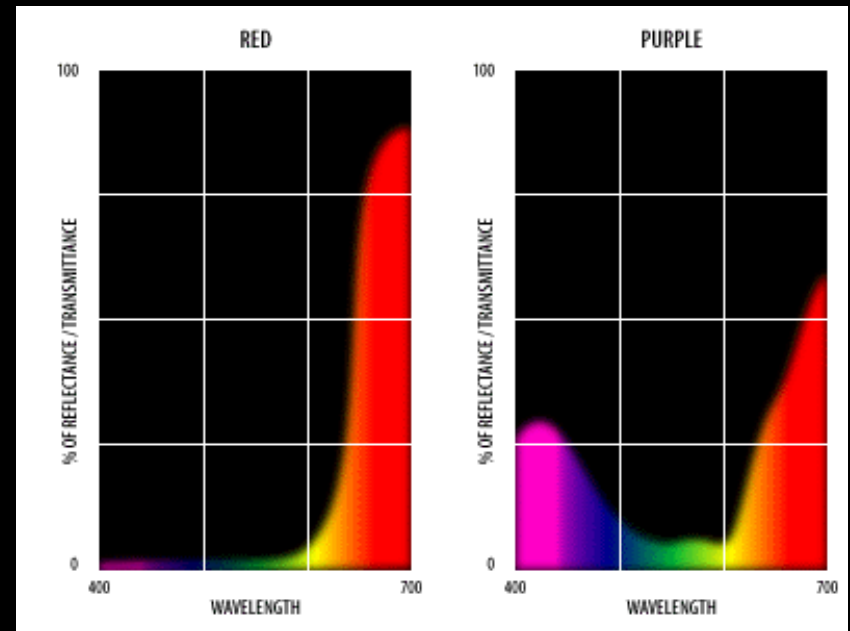
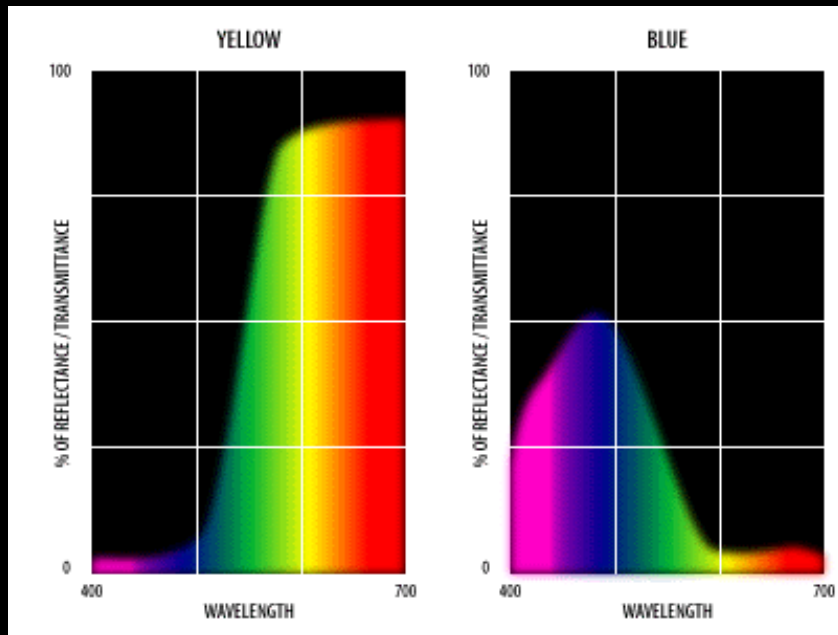
Color Perception

- When light strikes an object,
 - It will be wholly or partly transmitted.
 - It will be wholly or partly reflected.
 - It will be wholly or partly absorbed.
 - Physical surface properties dictate what happens
- When we see an object as blue or red or purple,
 - what we're really seeing is a partial reflection of light from that object.
 - The color we see is what's left of the spectrum after part of it is absorbed by the object.



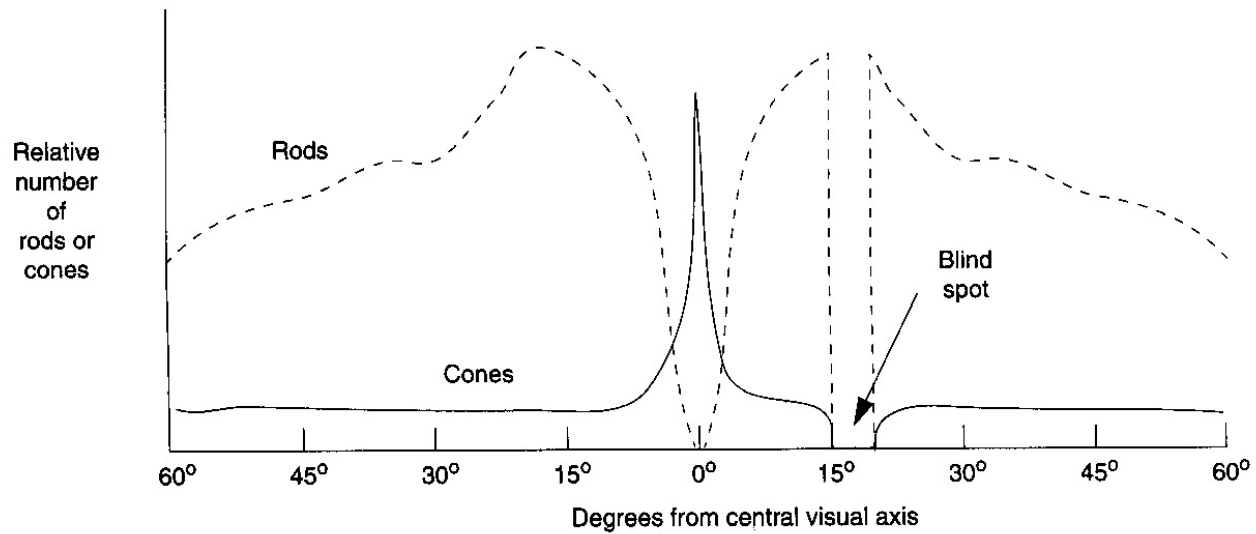
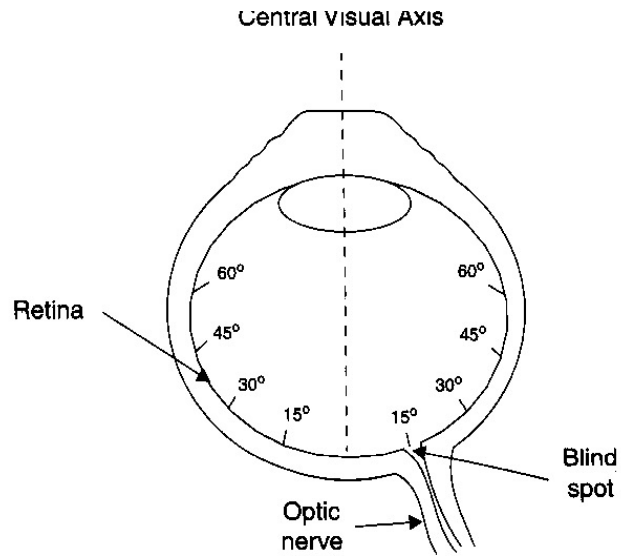
Spectral Reflectance Curves

- Reflectance curves for objects that appear to be:

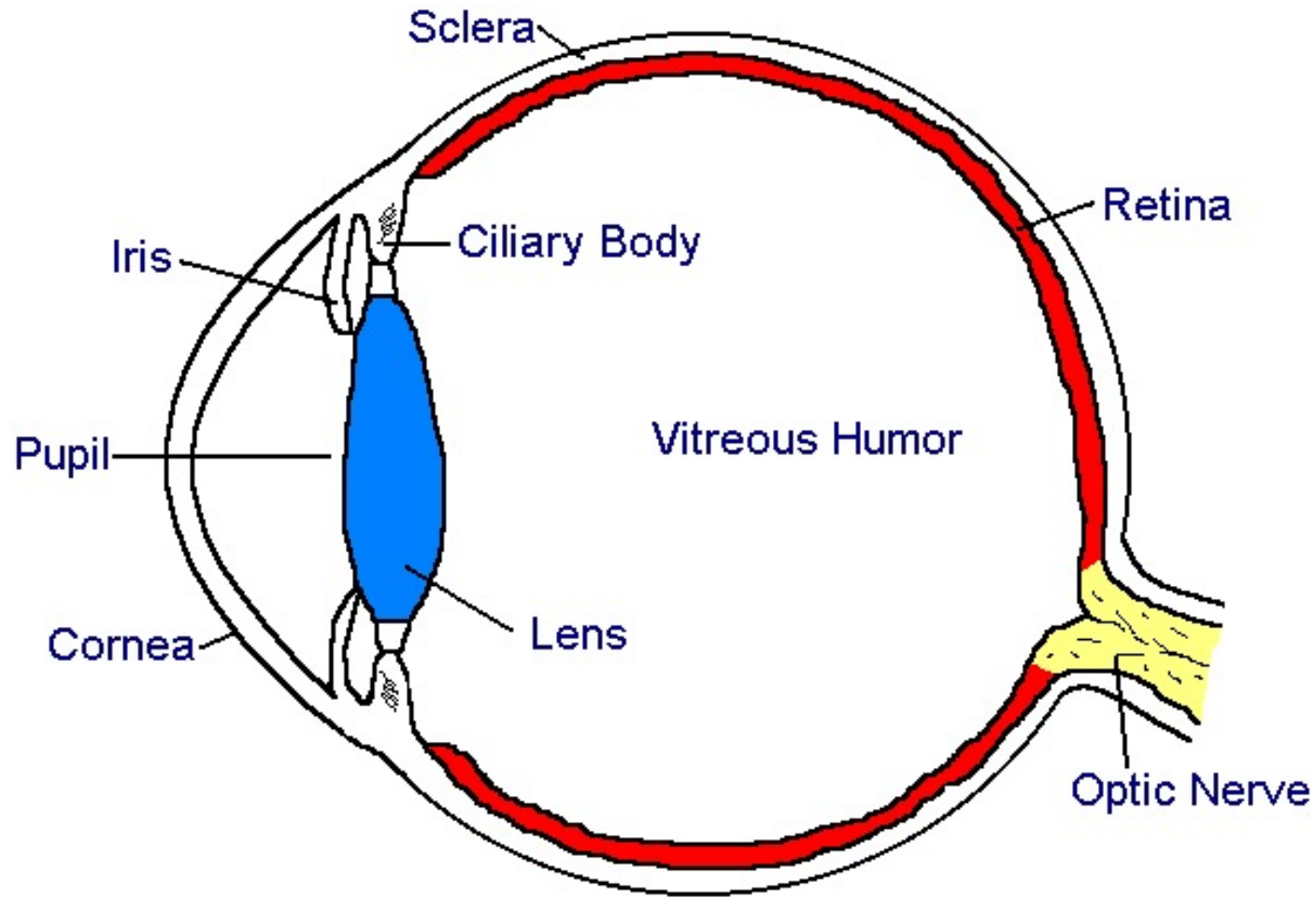


The wavelengths reflected or transmitted from or through an object determine the stimulus to the retina that provokes the optical nerve into sending responses to our brains that indicate color.

THE HUMAN EYE



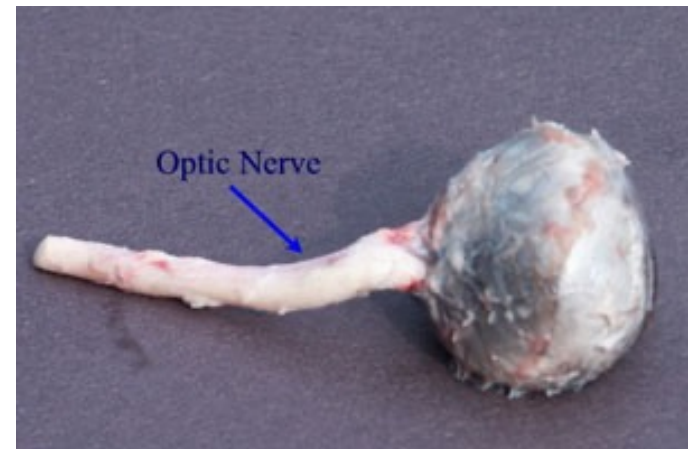
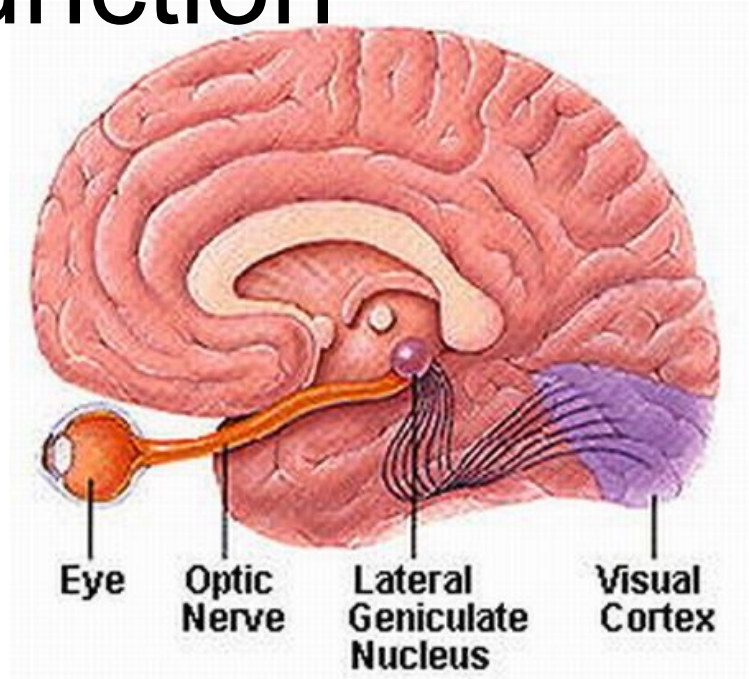
The Human Eye



Structure & Function

OPTIC NERVE

- Transmits electrical impulses from retina to the brain
- Creates blind spot
- Brain takes inverted image and flips it so we can see



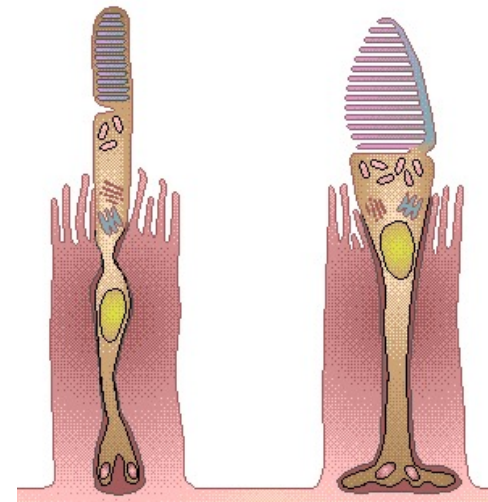
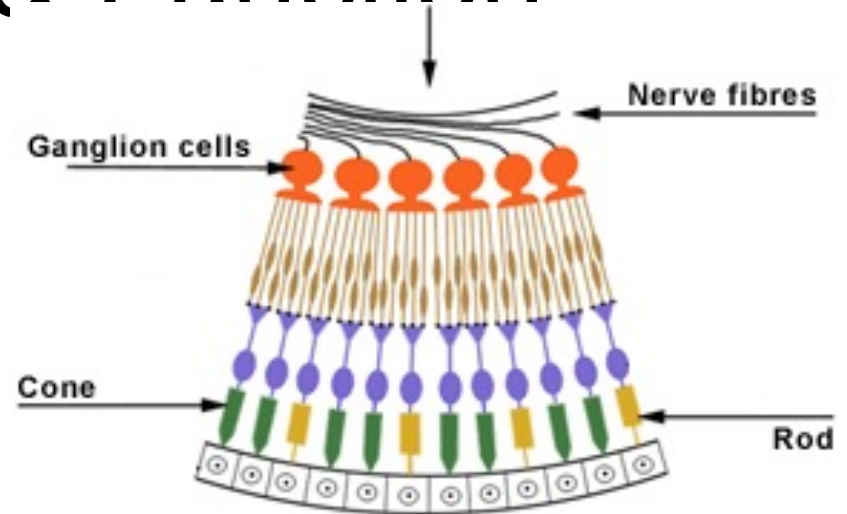
Structure & Function

- **RODS**

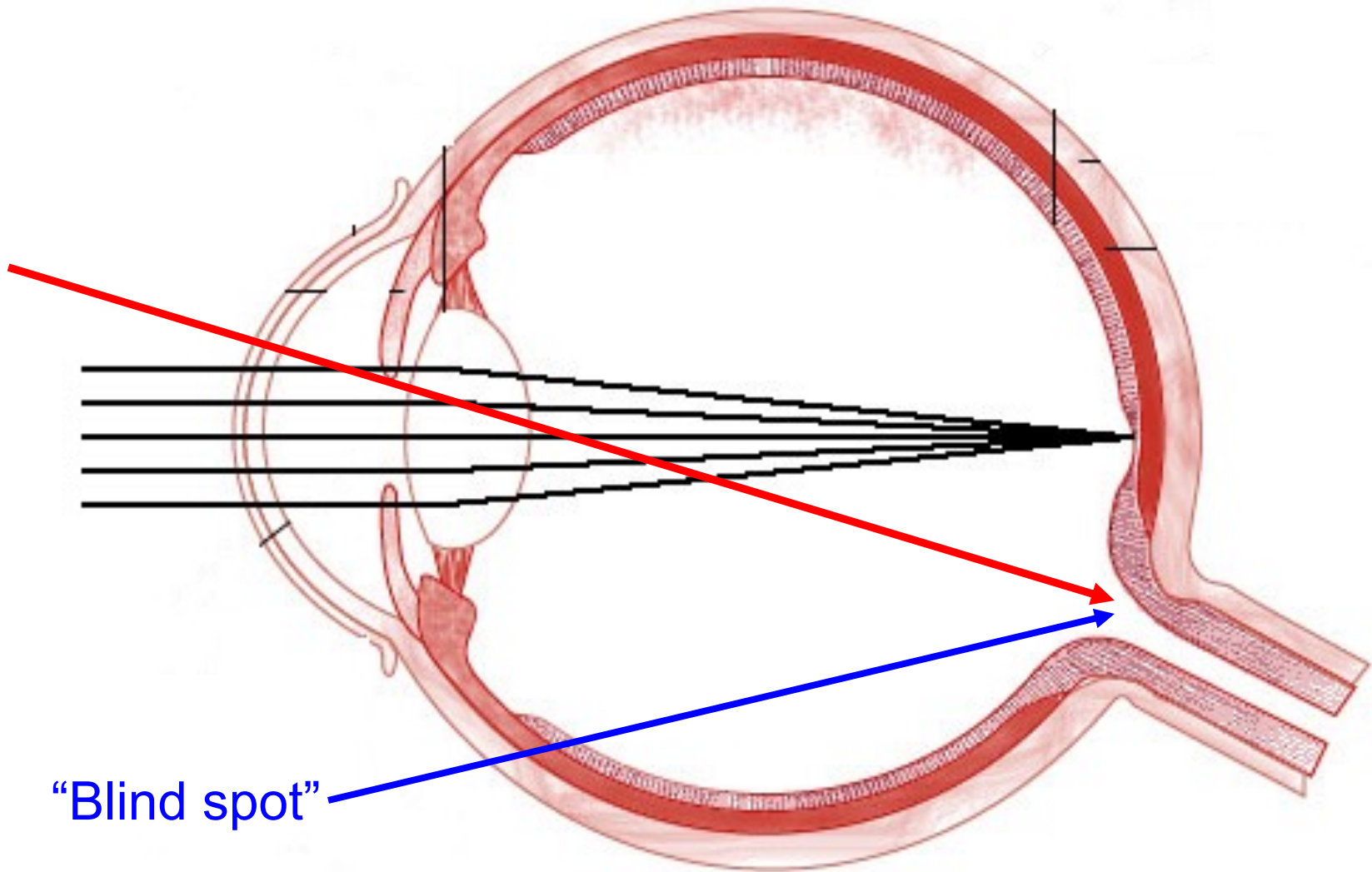
- 120 million cells
- detect brightness (black & white)
- for night vision

- **CONES**

- 6 million cells
- detect colour (RGB)
- GANGLION CELLS
- Detect movement and patterns



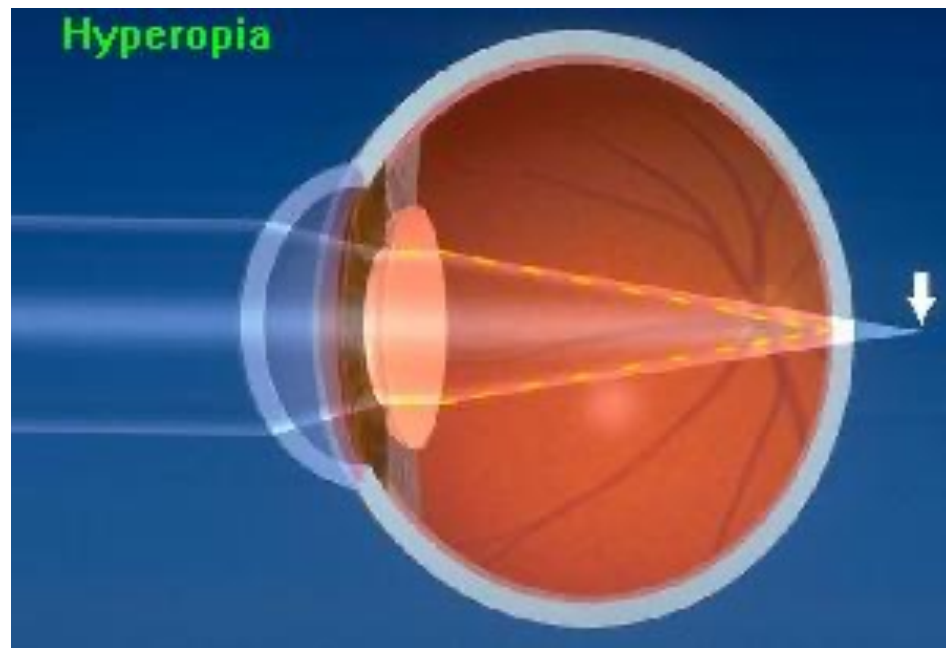
Normal Eye Focus



Focusing Problems

HYPEROPIA

- Far-sightedness
- Problem seeing close objects
- Distance between lens and retina too small
- Light focused behind retina
- Corrected with converging lenses



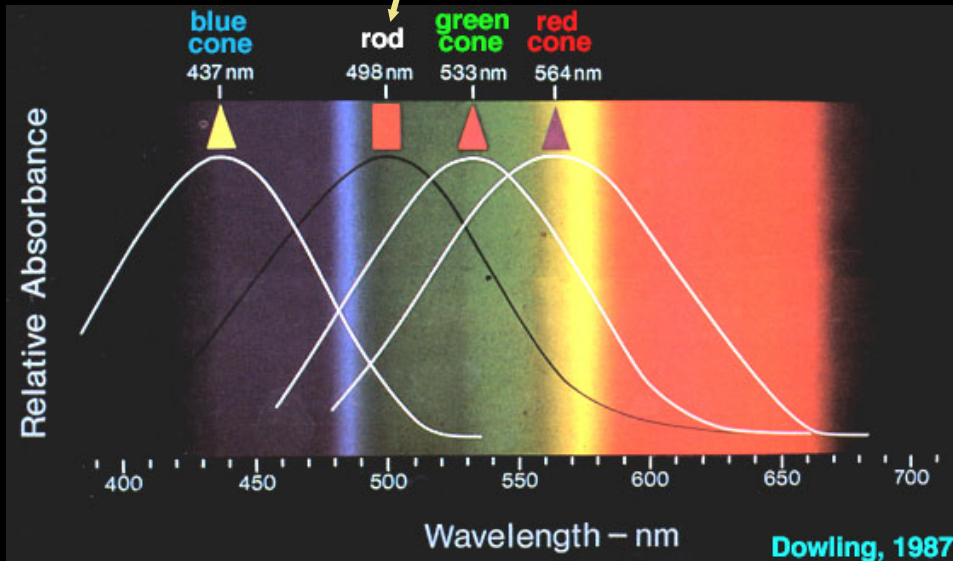
Eye Parts

- Rods – cylindrical photoreceptors found in greatest concentration on the edges of the retina; most common type of receptor; allow us to see in low light and provide for peripheral vision

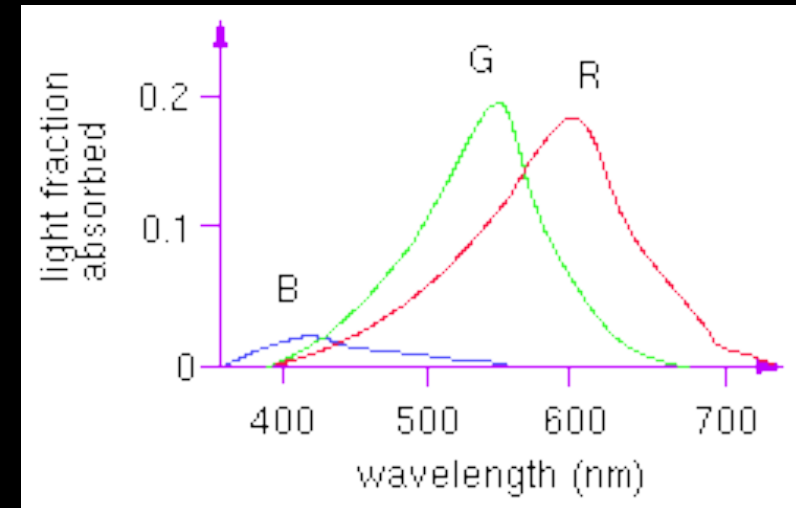
- Cones – Conical photoreceptors found in greatest concentration near the center of the retina; there are three varieties of cones, each most sensitive to a particular wavelength (color) of light – blue, green, & red; allow for visual acuity (sharp vision) and color vision

Absorption Curves

Rods: achromatic vision

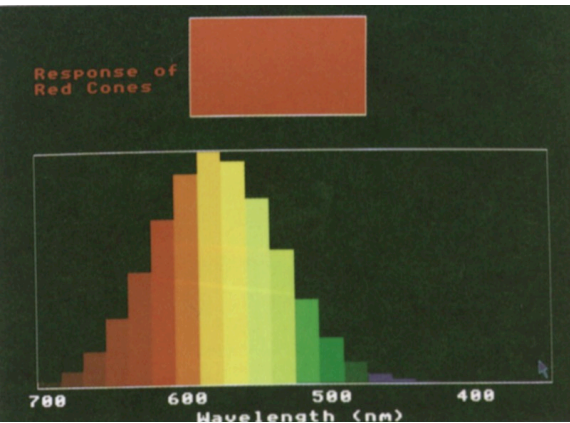


The different kinds of cells have different spectral sensitivities

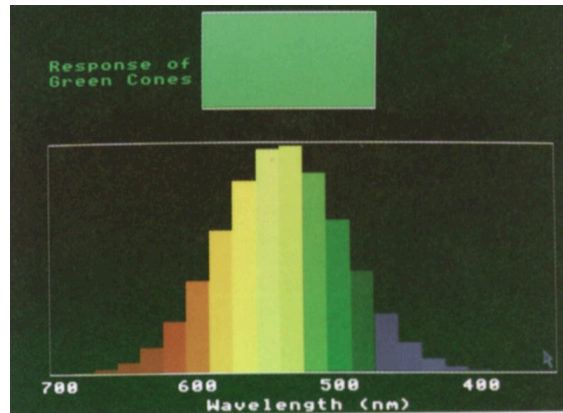


Peak sensitivities are located at approximately 437nm, 533nm, and 610nm for the "average" observer.

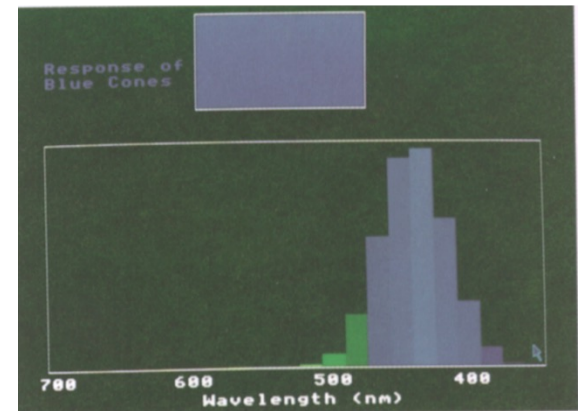
Color receptors



“Red” cone



“Green” cone

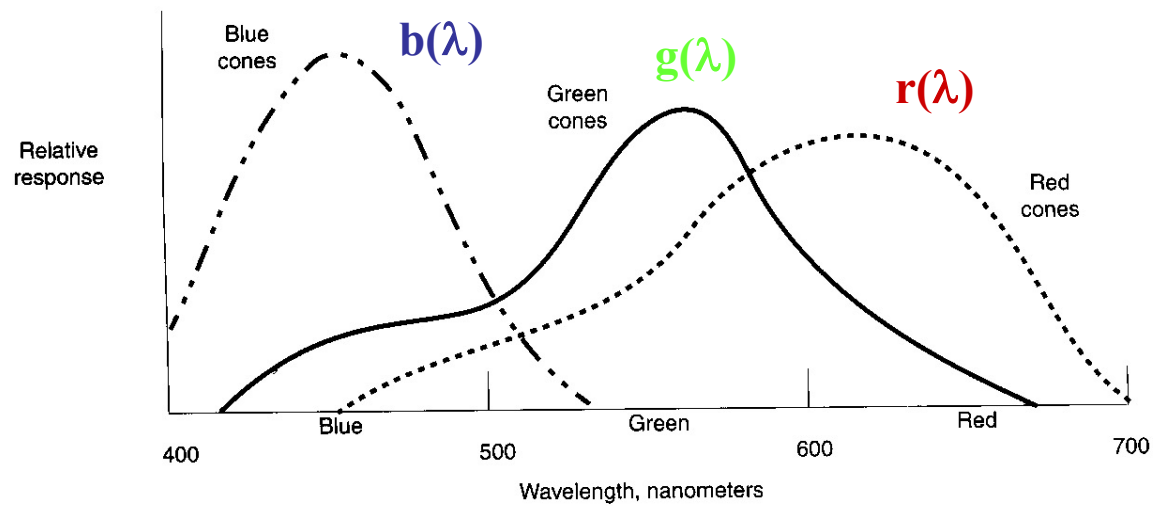


“Blue” cone

Principle of univariance: cones give the same kind of response, in different amounts, to different wavelengths. Output of cone is obtained by summing over wavelengths.

Responses measured in a variety of ways

$$\text{Response of } k\text{'th cone} = \int \rho_k(\lambda) E(\lambda) d\lambda$$



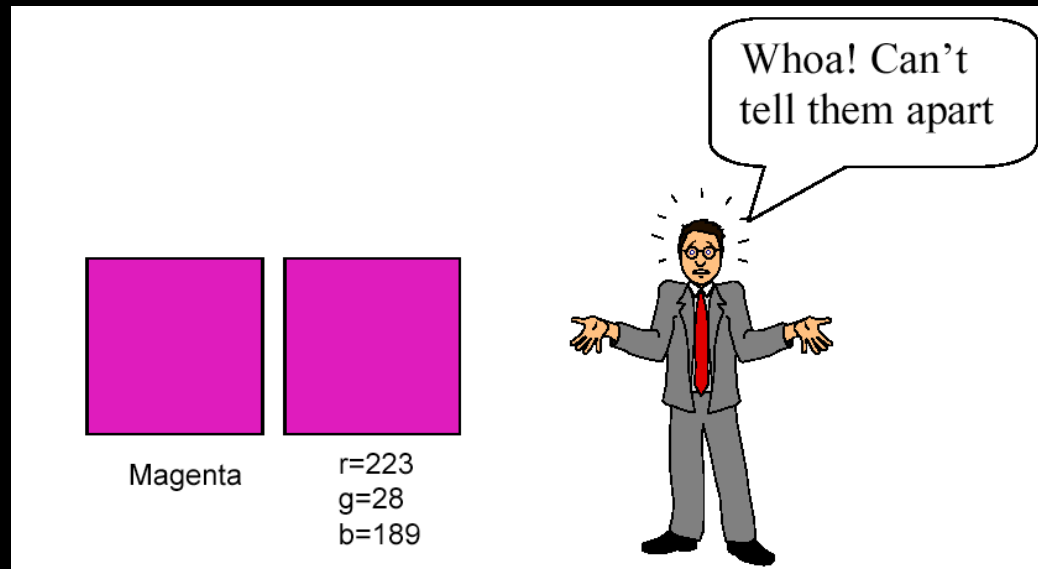
$$\text{RED} = \int_{\lambda=400nm}^{\lambda=700nm} r(\lambda)I(\lambda)d\lambda$$

$$\text{GREEN} = \int_{\lambda=400nm}^{\lambda=700nm} g(\lambda)I(\lambda)d\lambda$$

$$\text{BLUE} = \int_{\lambda=400nm}^{\lambda=700nm} b(\lambda)I(\lambda)d\lambda$$

Grassmann's Law (1835)

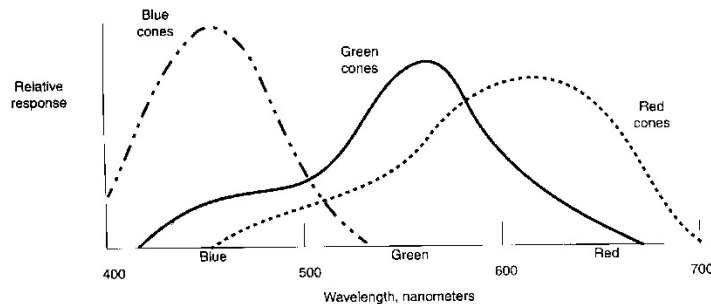
- 1st Law: Any color stimulus can be matched exactly by a combination of three primary lights.
 - The match is independent of intensity
- Basis of many color description systems



- 2nd Law: adding another light to both of these stimuli changes both in the same way.

METAMERISM

- Two different Spectral Energy Distributions with the same RED, GREEN, BLUE response are termed *metamers*.

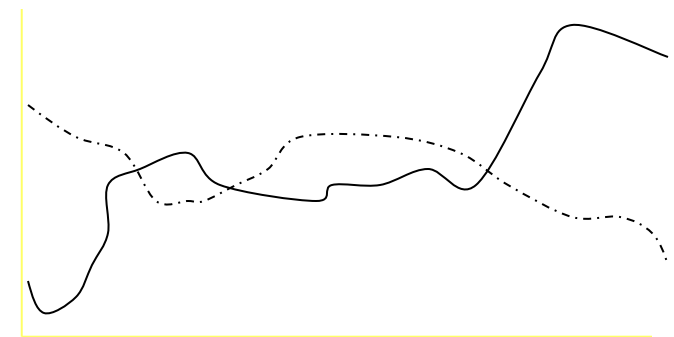


$b(\lambda)$

$g(\lambda)$

$r(\lambda)$

Radiance
(Energy)



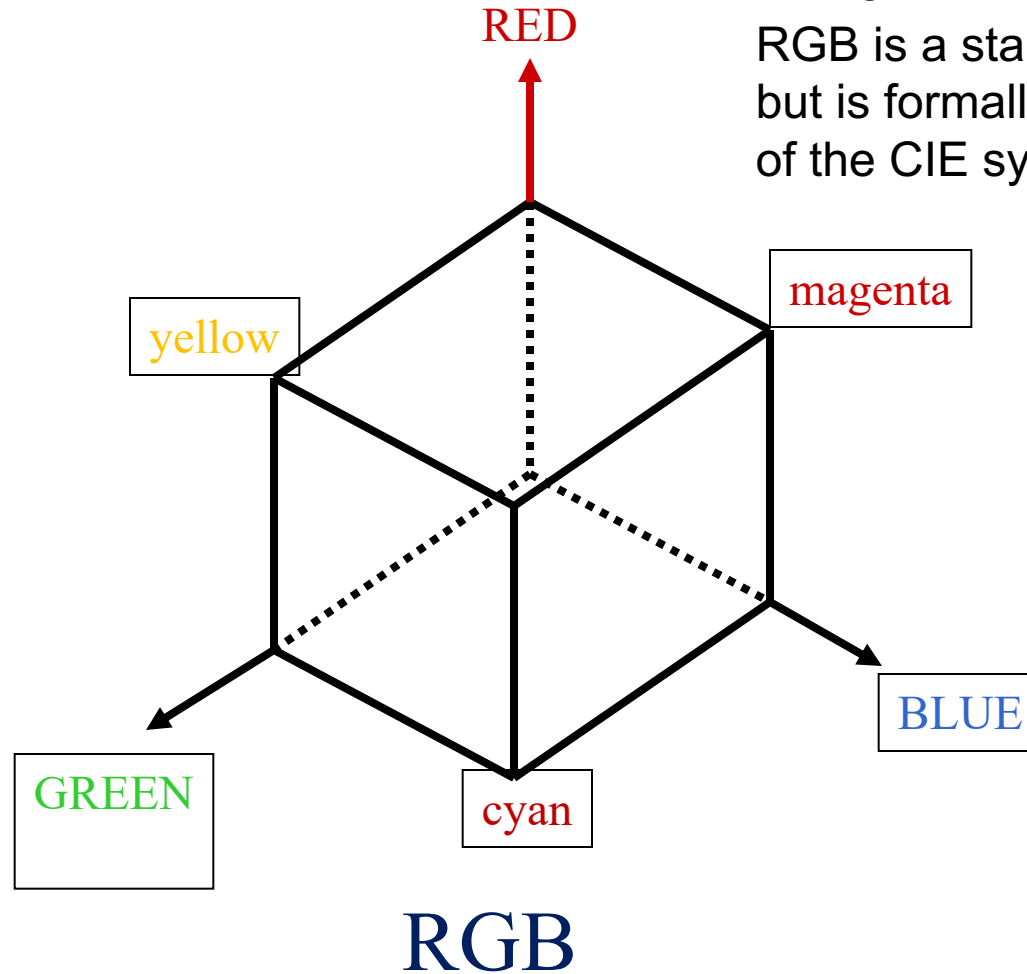
Wavelength
 λ

$$\text{RED} = \int_{\lambda=400nm}^{\lambda=700nm} r(\lambda)I(\lambda)d\lambda$$

$$\text{GREEN} = \int_{\lambda=400nm}^{\lambda=700nm} g(\lambda)I(\lambda)d\lambda$$

$$\text{BLUE} = \int_{\lambda=400nm}^{\lambda=700nm} b(\lambda)I(\lambda)d\lambda$$

Standard Color System



RGB is a standard system,
but is formally defined in terms
of the CIE system.

SCHEMES FOR REPRESENTING COLOR

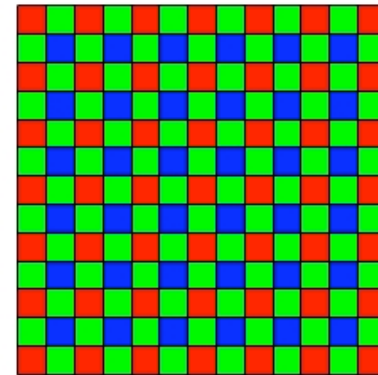
- Invented for color television
- Backward compatible with B/W TV
- Y given higher bandwidth than I/Q

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} .3 & .59 & .11 \\ .6 & -.28 & -.32 \\ .21 & -.52 & .31 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

YIQ (also YUV)

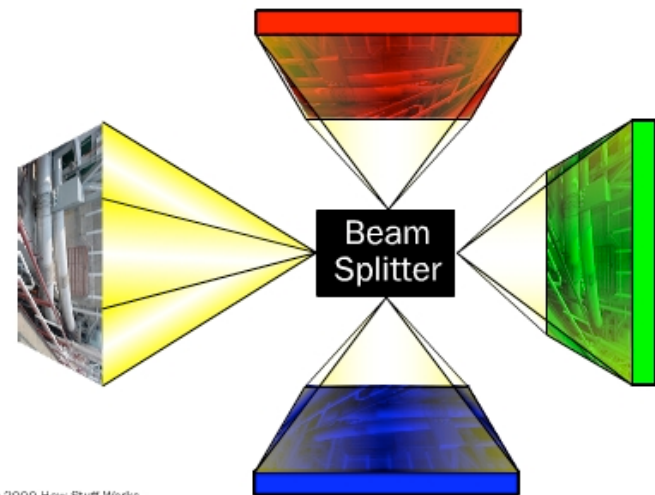
How Color Cameras Work

- 1 CCD cameras
 - A **Bayer** pattern is placed in front of the CCD
 - A **Demosaicing** process reads the pixels in a region and computes color and intensity
- 3 CCD camera use a beam splitter and 3 separate CCDs
 - higher color fidelity
 - needs lots of light
 - requires careful alignment of ccDs



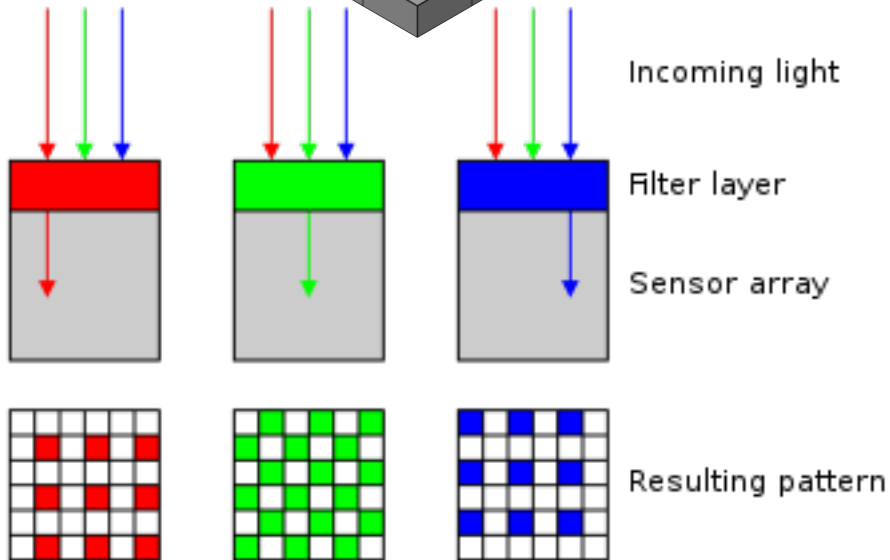
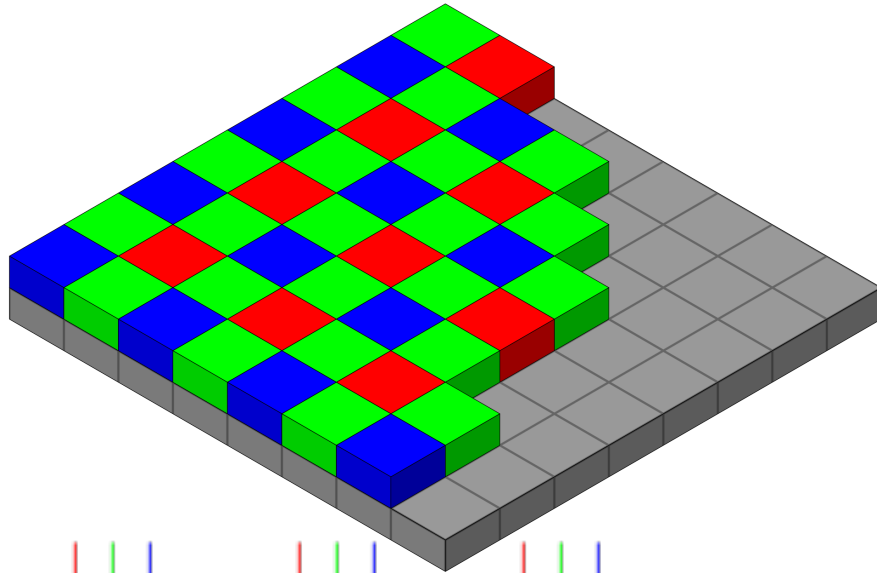
Bayer filter

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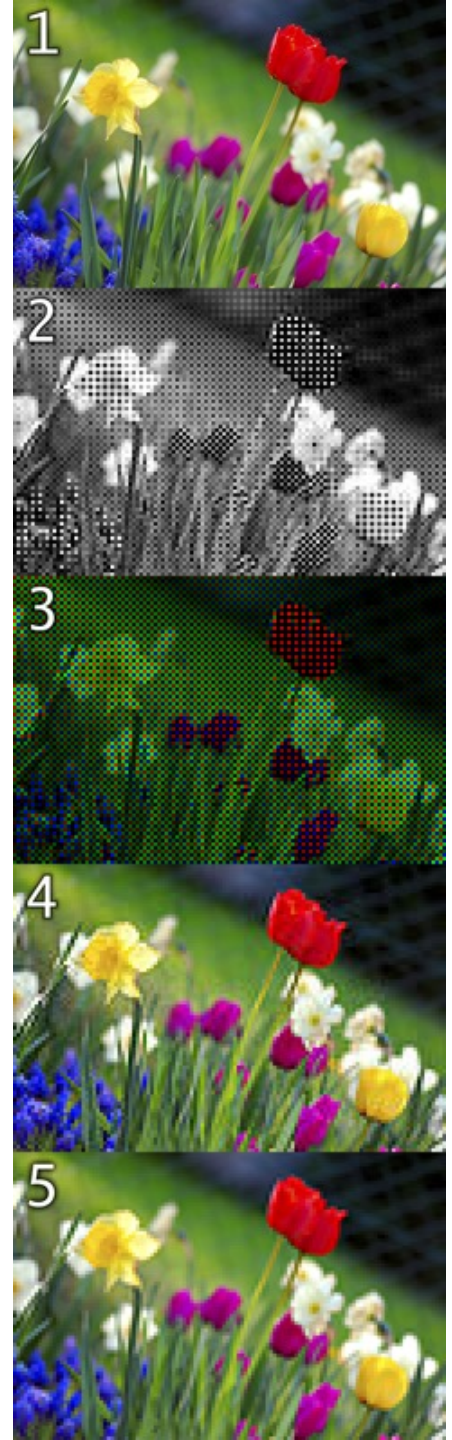


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Raw Image Format and Bayer Filter



1. Original scene
2. Original scene Output of a 120×80-pixel sensor with a Bayer filter
3. Output color-coded with Bayer filter colors
4. Reconstructed image after interpolating missing color information
5. Full RGB version at 120×80-pixels



Dividing Up Color Space

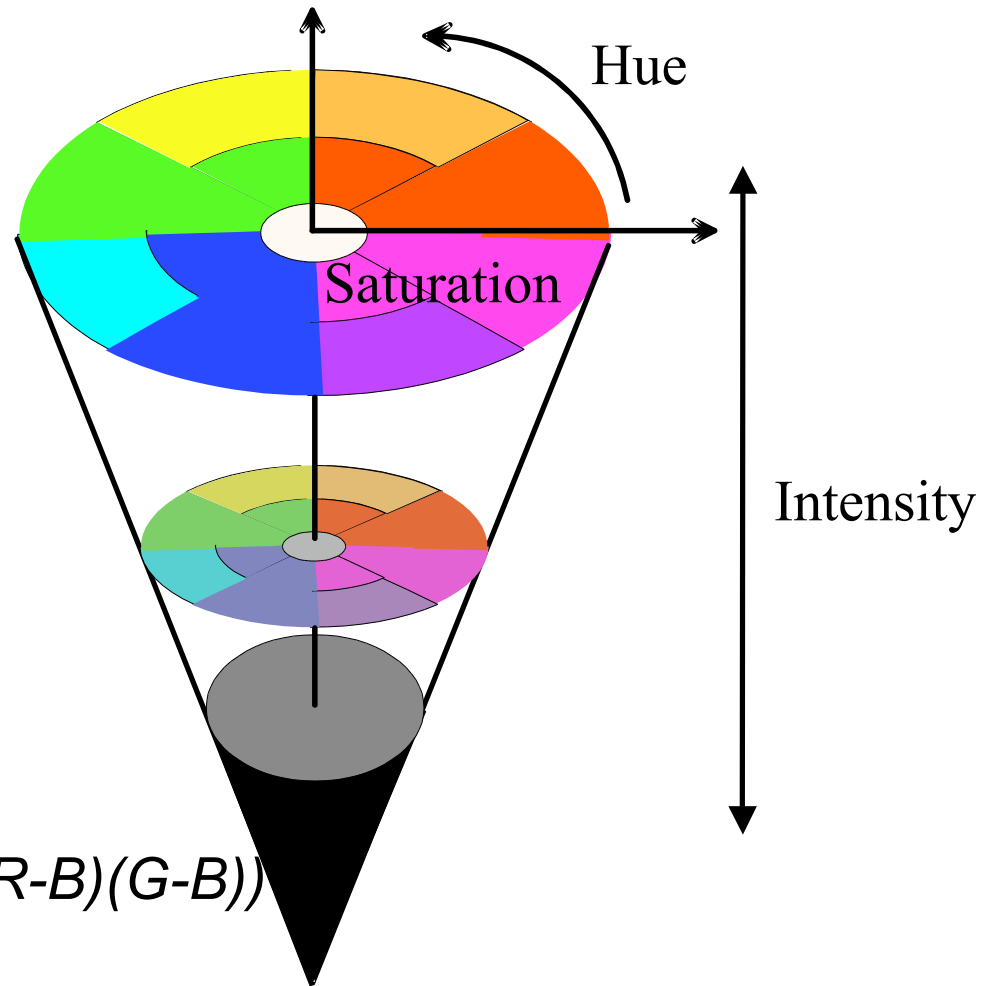
HSI is a nonlinear representation of color space. Note the non-uniform treatment of color

$$I = (R+G+B)/3 \text{ or} \\ L = .3 R + .6 G + .1 B$$

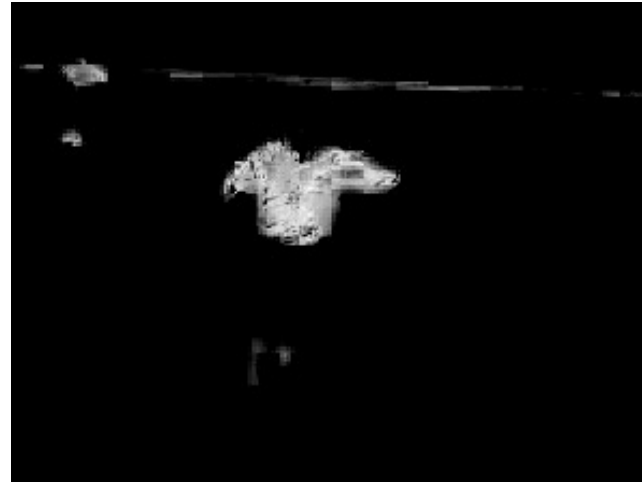
$$S = 1 - 3 \min(R, G, B)/I$$

$$H = \begin{cases} \cos^{-1}(x) & \text{if } G > B \\ \pi - \cos^{-1}(x) & \text{if } G < B \end{cases}$$

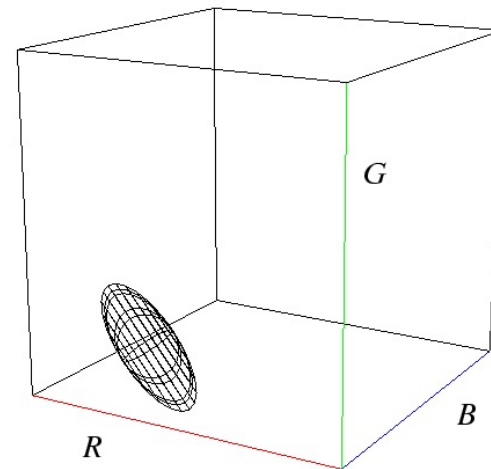
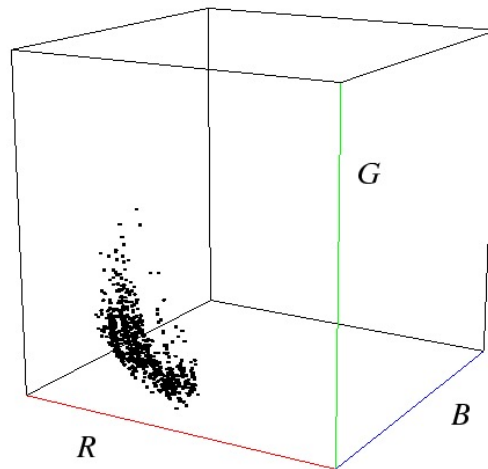
$$x = (R-G) + (R-B)/((R-G)^2 + (R-B)(G-B))$$



Homogeneous Region: Photometry

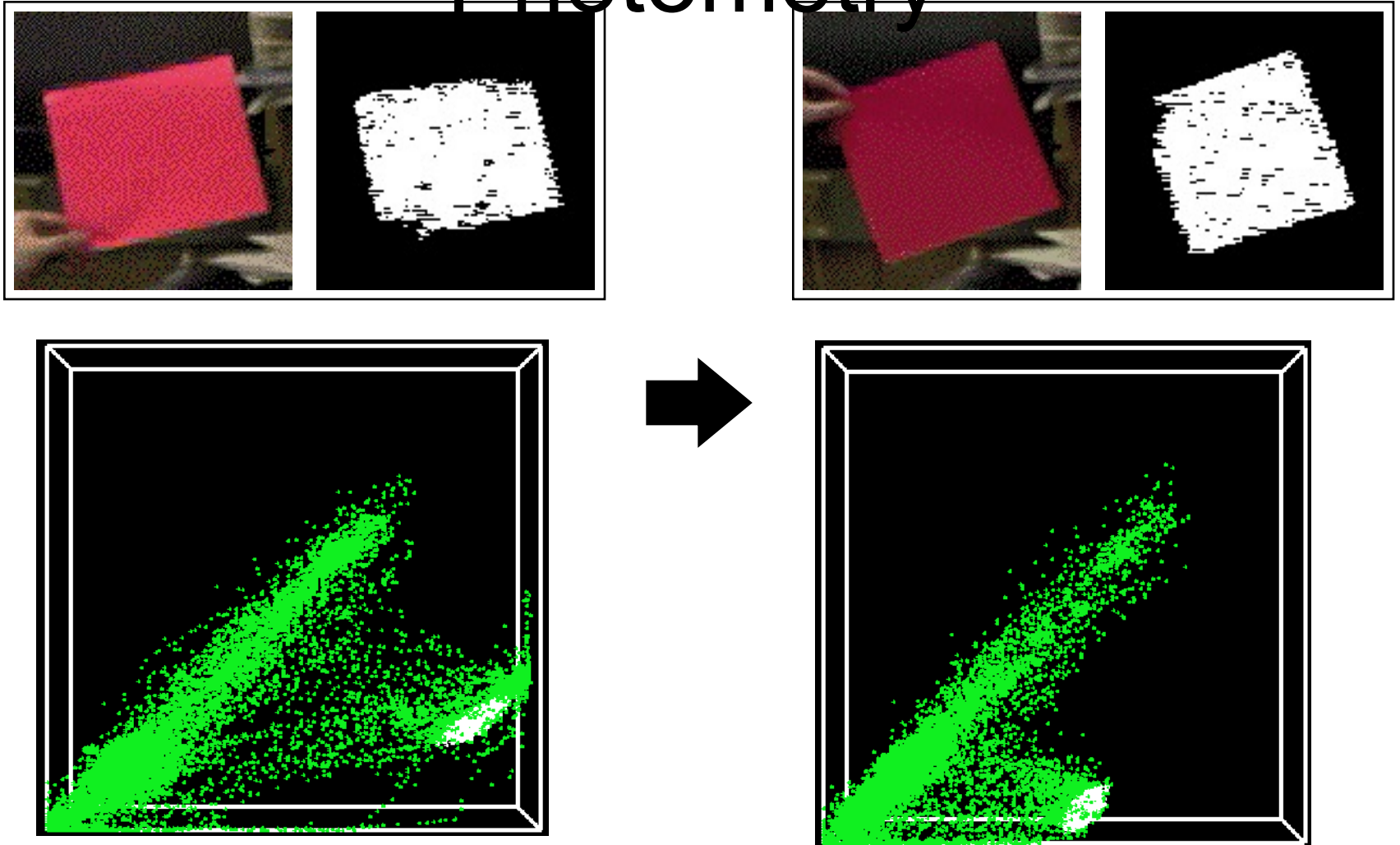


Sample



PCA-fitted
ellipsoid

Homogeneous Color Region: Photometry



Finding Specularities

- Assume we are dealing with dielectrics
 - specularly reflected light is the same colour as the source
- Reflected light has two components
 - diffuse
 - specular
 - and we see a weighted sum of these two

